

BodySimPy Engineering Summary

Objective

Automate structural CAE workflows for a simplified automotive body crossmember surrogate using Python, analytical references, CalculiX FEA, stochastic studies, fatigue assessment and ML surrogates.

Method

Validated configuration files generate structural models. Analytical beam theory, CalculiX static/modal analysis and Python post-processing are used to quantify response behavior.

Baseline Configuration

Thin-walled rectangular hollow-section cantilever surrogate with linear-elastic isotropic material and idealized transverse loading.

Key Results

Final mesh displacement error: 0.055 %

Final mesh stress error: 4.641 %

Mode-1 frequency: 49.401 Hz

Observed stochastic stress exceedance over 350 MPa: 0.000 %

Best ML frequency MAPE: 0.242 %

Sensitivity

Wall thickness, section geometry, Young's modulus, density and load are evaluated through automated sweeps and stochastic analysis.

Risk

The model is a simplified surrogate, not a production body-in-white. Results depend on idealized boundary conditions, beam assumptions, generic material data and the sampled design domain.

Engineering Recommendation

Use BodySimPy as a reproducible CAE automation and validation portfolio workflow. Extend shell modelling, fatigue realism and solver-result extraction before claiming production-level accuracy.